

Physical Parameters Corresponding to the Chaotic Dimensionless State

Reverse calculation and proof of the conditions $\varepsilon_1 = 0$ and $\varepsilon_3 = 0$

The equation numbers refer to the revised paper:

- (3): physical model;
- (4): coefficients;
- (5): dimensionless model;
- (6): definition of γ .

The chaotic set is $\gamma = -1$, $s = 8$, $\varepsilon_1 = 0$, $\varepsilon_2 = -85$, $\varepsilon_3 = 0$. All other constants are taken from Table I (stator-referred 2-MW-class DFIG).

1 Model and scaling

The physical model (3) with $u = 0$ is

$$\begin{cases} \dot{x}_1 = a_1 x_1 + (\omega_s - x_3)x_2 + a_2 x_3 + c_1 \\ \dot{x}_2 = -(\omega_s - x_3)x_1 + a_1 x_2 + c_2 \\ \dot{x}_3 = a_6 x_1 - a_7 x_3 - a_8 \end{cases}$$

with the coefficients

$$\begin{aligned} a_1 &= -\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r}, & a_2 &= -\frac{L_m \psi_s}{\sigma L_r L_s}, & a_3 &= -\frac{L_m}{\sigma L_s L_r}, & a_4 &= \frac{1}{\sigma L_r}, & a_5 &= \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r}, \\ a_6 &= \frac{3p^2 L_m \psi_s}{2J L_s}, & a_7 &= \frac{D}{J}, & a_8 &= \frac{p}{J} T_L, & c_1 &= a_3 u_{ds} + a_4 u_{dr}^0, & c_2 &= a_5 + a_3 u_{qs} + a_4 u_{qr}^0. \end{aligned}$$

The change of variables used to obtain (5) is

$$t = \tau \tilde{t}, \quad x_{1,2} = k z_{1,2}, \quad x_3 = \omega_s + \frac{z_3}{\tau}, \quad \tau = -\frac{1}{a_1}, \quad k = \frac{a_7}{\tau a_6}.$$

2 Derivation of the dimensionless parameters

First equation. Use $\omega_s - x_3 = -z_3/\tau$ and $a_2 x_3 = a_2 \omega_s + a_2 z_3/\tau$, then multiply by τ/k :

$$z_1' = \frac{\tau}{k} \dot{x}_1 = \underbrace{\tau a_1}_{-1} z_1 - z_3 z_2 + \underbrace{\frac{a_2}{k}}_{\gamma} z_3 + \underbrace{\frac{\tau}{k} (c_1 + a_2 \omega_s)}_{\varepsilon_1}.$$

Second equation.

$$z_2' = -z_2 + z_3 z_1 + \underbrace{\frac{\tau}{k} c_2}_{\varepsilon_2}.$$

Third equation. Here $z_3' = \tau^2 \dot{x}_3$, so

$$z_3' = \underbrace{\tau^2 a_6 k}_{s} z_1 - \underbrace{\tau a_7}_{s} z_3 - \underbrace{\tau^2 (a_7 \omega_s + a_8)}_{\varepsilon_3}.$$

Write $a_6 = c_6/J$ with $c_6 = \frac{3p^2 L_m \psi_s}{2L_s}$. Then J cancels out of γ and k :

$$\gamma = \frac{a_2 a_6 \tau}{a_7} = \frac{a_2 c_6 \tau}{D} = -\frac{3p^2 L_m^2 \psi_s^2 \tau}{2\sigma L_r L_s^2 D} < 0, \quad k = \frac{D}{\tau c_6} = \frac{a_2}{\gamma}.$$

3 Inverse relations

Inverting the previous relations gives the five physical unknowns:

$$\boxed{D = \frac{a_2 c_6 \tau}{\gamma}}, \quad \boxed{J = \frac{\tau D}{s}}, \quad \boxed{u_{dr}^0 = \frac{L_m}{L_s} (u_{ds} + \omega_s \psi_s) + \frac{\varepsilon_1 k}{\tau a_4}},$$

$$\boxed{u_{qr}^0 = \frac{1}{a_4} \left(\frac{\varepsilon_2 k}{\tau} - a_5 - a_3 u_{qs} \right)}, \quad \boxed{T_L = -\frac{J}{p} \left(\frac{\varepsilon_3}{\tau^2} + \frac{D}{J} \omega_s \right)}.$$

4 Numerical values

Fixed constants (Table I). They depend only on the electrical parameters, not on J or D :

- $\sigma = 0.06613$, $\tau = 0.032108$ s, $\psi_s = 1.7933$ Wb, $\omega_s = 314.16$ rad/s;
- $a_2 = -10130.07$ A/rad, $a_4 = 5845.42$ A/(V·s), $a_5 = 10180.98$ A/s;
- $c_6 = 10.398$ N·m/A.

Parameter	Calculation	Chaotic value	Nominal / limit	Ratio
D	$a_2 c_6 \tau / \gamma$	3382.0 N·m·s/rad	10^{-3}	$\times 3.4 \cdot 10^6$
J	$\tau D / s$	13.57 kg·m ²	127	$\div 9.4$
D/J	s / τ	249.2 s ⁻¹	$7.9 \cdot 10^{-6}$	$\times 3.2 \cdot 10^7$
k	a_2 / γ	10130 A	—	—
u_{dr}^0	$\varepsilon_1 = 0$	0 V	—	—
u_{qr}^0	$(\varepsilon_2 k / \tau - a_5) / a_4$	-4589.5 V	≈ 200 V	$\times 23$
T_L	$-D \omega_s / p$	-531.2 kN·m	12.7 kN·m rated	$\times 42$

5 Proof of the conditions $\varepsilon_1 = 0$ and $\varepsilon_3 = 0$

Since $\tau/k \neq 0$ and $\tau^2 \neq 0$, Section 2 gives

$$\varepsilon_1 = 0 \iff c_1 + a_2 \omega_s = 0, \quad \varepsilon_3 = 0 \iff a_7 \omega_s + a_8 = 0.$$

5.1 Condition $\varepsilon_1 = 0$: rotor d -axis voltage

Using $c_1 = a_3 u_{ds} + a_4 u_{dr}^0$:

$$a_3 u_{ds} + a_2 \omega_s = -\frac{L_m}{\sigma L_s L_r} u_{ds} - \frac{L_m \psi_s}{\sigma L_r L_s} \omega_s = -\frac{L_m}{\sigma L_s L_r} (u_{ds} + \omega_s \psi_s).$$

Hence

$$\varepsilon_1 = 0 \iff \frac{u_{dr}^0}{\sigma L_r} = \frac{L_m}{\sigma L_s L_r} (u_{ds} + \omega_s \psi_s) \iff \boxed{u_{dr}^0 = \frac{L_m}{L_s} (u_{ds} + \omega_s \psi_s)}.$$

- **With the paper's assumption.** The stator resistance is neglected in the flux relation and the flux is aligned on the q axis, so $u_{ds} = -\omega_s \psi_s = -V_s$. Then

$$\boxed{u_{dr}^0 = 0 \text{ V}}.$$

Numerically, $a_3 u_{ds} = +3.1825 \times 10^6 \text{ A/s}$ and $a_2 \omega_s = -3.1825 \times 10^6 \text{ A/s}$ cancel exactly.

- **Physical meaning.** The term $a_2 x_3$ is the back-EMF of the stator flux seen by the rotor. At synchronous speed it is exactly compensated by the stator voltage, so no rotor d -axis voltage is needed. The condition is purely electrical: it does not depend on J , D or T_L .
- **If R_s is kept.** Then $u_{ds} = -\omega_s \psi_s + R_s i_{ds}$, and $\varepsilon_1 = 0$ requires $u_{dr}^0 = \frac{L_m}{L_s} R_s i_{ds}$, which is only a few mV per kA. In all cases, $\varepsilon_1 = 0$ means an essentially zero rotor d -axis voltage.

5.2 Condition $\varepsilon_3 = 0$: load torque

With $a_7 = D/J$ and $a_8 = pT_L/J$:

$$a_7 \omega_s + a_8 = \frac{D\omega_s + pT_L}{J} = 0 \iff \boxed{T_L = -\frac{D\omega_s}{p} = -D\Omega_s},$$

where $\Omega_s = \omega_s/p = 157.08 \text{ rad/s}$ is the synchronous mechanical speed.

- **Dependence.** J cancels: the condition depends only on D . With the motor convention, $T_L < 0$ means that the turbine *drives* the shaft.
- **Physical meaning.** For $x_1 = 0$ (zero electromagnetic torque) and $x_3 = \omega_s$, the third line of (3) gives $\dot{x}_3 = -a_7 \omega_s - a_8 = 0$. Thus $\varepsilon_3 = 0$ means that synchronous speed is an equilibrium of the mechanical equation without electromagnetic torque: the driving torque exactly compensates the viscous loss $D\Omega_s$.

Case	D (N·m·s/rad)	Required T_L	vs. rated (12.7 kN·m)
Nominal (Table I)	10^{-3}	-0.157 N·m	negligible (almost no load)
Chaotic set ($\gamma = -1$, $s = 8$)	3382	-531.2 kN·m	42 times rated

The condition $\varepsilon_3 = 0$ is harmless in itself. The required torque becomes impossible only because $\gamma = -1$ imposes a huge D .

5.3 Why choose $\varepsilon_1 = \varepsilon_3 = 0$: symmetry

With $\varepsilon_1 = \varepsilon_3 = 0$, system (5) is invariant under the transformation

$$(z_1, z_2, z_3) \mapsto (-z_1, z_2, -z_3).$$

Line-by-line check:

- first line: $-z'_1 = z_1 + z_3 z_2 - \gamma z_3$, which gives back $z'_1 = -z_1 - z_3 z_2 + \gamma z_3$;
- second line: $z'_2 = -z_2 + (-z_3)(-z_1) + \varepsilon_2$;
- third line: $-z'_3 = s(-z_1 + z_3)$.

If $\varepsilon_1 \neq 0$ or $\varepsilon_3 \neq 0$, a constant term changes sign and the symmetry is broken. This explains the symmetric double-scroll attractor of Fig. 1(a)–(b), and leaves ε_2 as the only bifurcation parameter.

6 Role of each dimensionless parameter

- $\gamma = -1$ **fixes D alone**, since $\gamma = a_2 c_6 \tau / D$ does not contain J . Its sign is imposed by physics: $a_2 < 0$ whenever $\sigma > 0$. With the nominal $D = 10^{-3}$, one would have $\gamma \approx -3.4 \times 10^6$.
- $s = 8$ **then fixes $J = \tau D / s$** . It means that the mechanical time constant $J/D = 4$ ms is 8 times shorter than the electrical time constant $\tau = 32$ ms. In a real turbine, the mechanical time constant is of the order of seconds to minutes.
- $\varepsilon_2 = -85$ **fixes $u_{qr}^0 = -4.59$ kV**. The bifurcation diagram shows chaos for $\varepsilon_2 < -15$, i.e. $u_{qr}^0 < -812$ V. Even the onset of chaos is about four times the converter limit.

7 Conclusion

Quantity	Chaotic set	Physical limit / nominal
Rotor q -axis voltage	−4.6 kV	≈ 200 V converter limit ($\times 23$); 8 times $V_s = 563$ V
Driving torque	531 kN·m	12.7 kN·m rated ($\times 42$)
Damping D	3382 N·m·s/rad	10^{-3} (6 orders of magnitude above)
Inertia J	13.6 kg·m ²	127 kg·m ² (9 times smaller)
Rotor currents on the attractor ($ z \approx 45$)	≈ 456 kA	≈ 2 kA rated ($\times 200$)
Speed on the attractor ($\omega_s \pm 25/\tau$)	−464 to +1093 rad/s	speed would even change sign

The chaotic state does not correspond to a normal operating condition. It is a severe fault/detuning scenario that is physically unreachable for a real 2-MW DFIG: the converter, the insulation and the protections would fail long before.

- $\varepsilon_1 = 0$ is physically realistic: zero rotor d -axis voltage at synchronous speed.
- $\varepsilon_3 = 0$ is realistic in form, but with the damping imposed by $\gamma = -1$ it requires a driving torque 42 times the rated torque.

This is consistent with the revised paper (Section III-B). It justifies studying chaos suppression on the dimensionless model (5), and MPPT tracking in physical units.